

Motivation

We want to see how well we could get a model to perform at Geoguessr. We all enjoy the game, but we are quite poor at it. We hoped a combination of DL and CV techniques could do better.



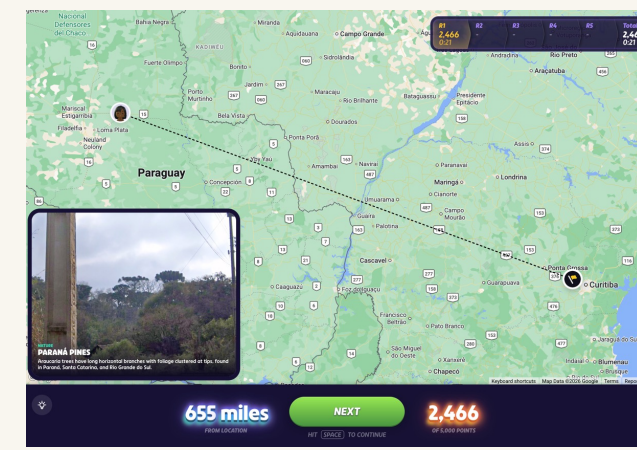
Problem

Feature Detection

The trick to Geoguessr is narrowing down the location from specific details in the image. Deep learning models are a great way to do this, storing quantities of info humans never could. Previous research had success with pretrained models like CLIP [1], so we decided to experiment with the state of the art DINOv3 [2] and also develop our own more classical approaches.

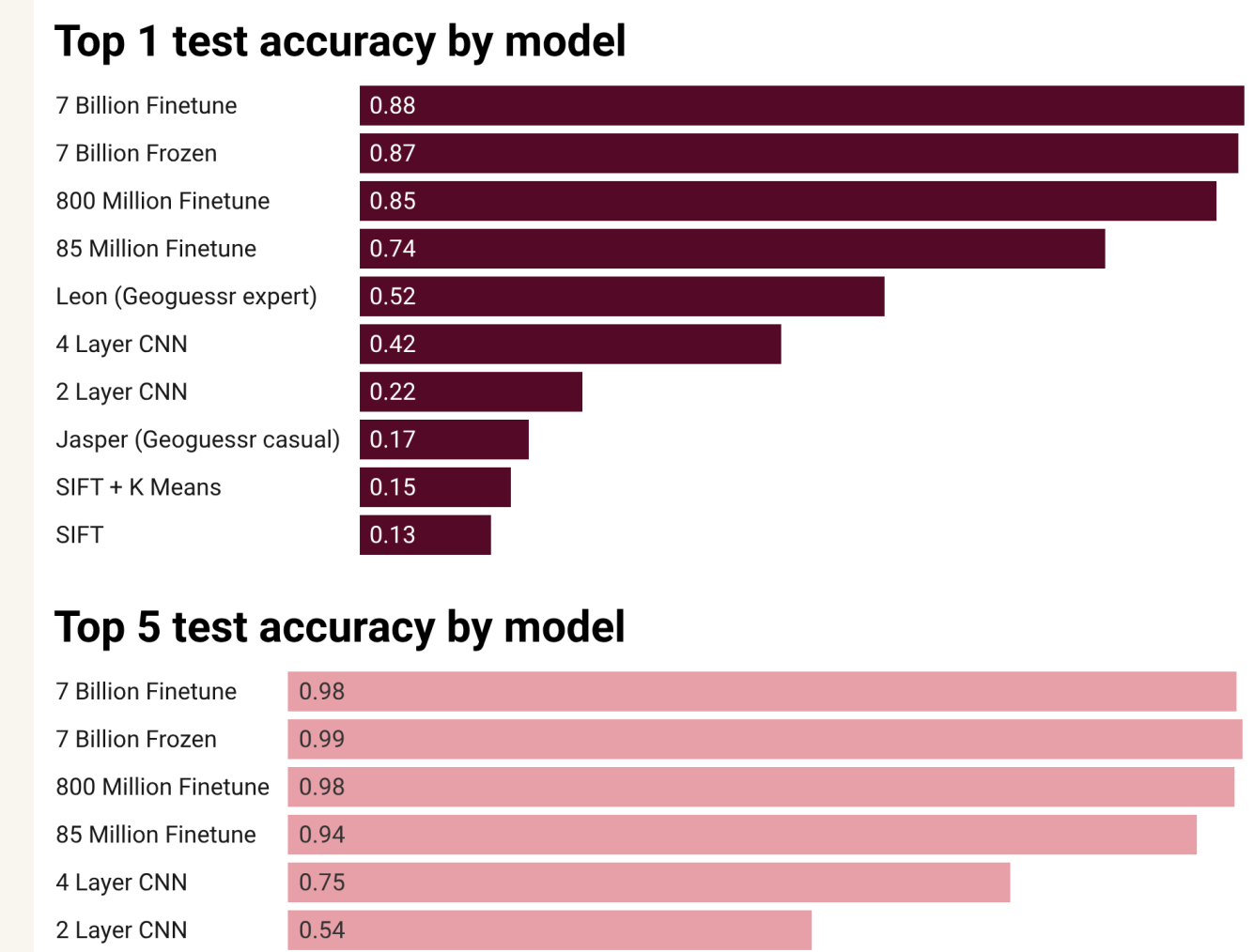
Localization/Classification

To actually play Geoguessr, our model would need to take the features it detects and turn them into a country/city classification or specific latitude/longitude coordinates. We experimented with both.



Goal

1. Outperform (good) humans on the same task
2. Build many different models to compare/build an ensemble
3. Better understand the strengths of the various CV/DL approaches



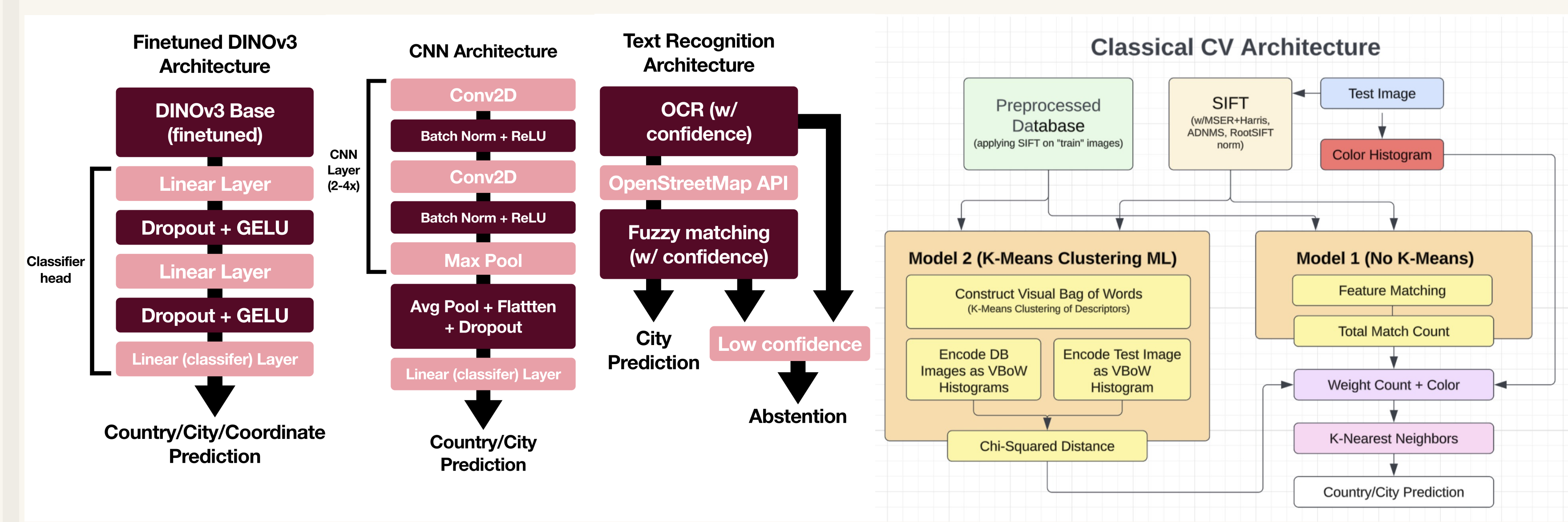
Insights

- As always in DL, need lots of data.
- Real Geoguessr players memorize thousands (millions?) of details.
- Mainly used a geoguessr specific database, but we also experimented with other datasets for coordinate and city data.
- The DINOv3 base was the most effective, but other methods also performed well.
- To have the computer perform like a human, it should "think" like a human. Classical CV tools like SIFT-like feature matching & histograms allow us to spot recurring objects/corners/themes that may be tied to certain countries, giving us better info to guess with.
- However, limitations arise quickly. It's slow, prone to noise, & SIFT wasn't meant for this (it's meant for 2 different images of the same scene) as seen with the invariances. Histograms can abstract too much of the patterns we'd need. It can spot pixel-to-pixel patterns, but it can't understand the "real" pattern, so to say.

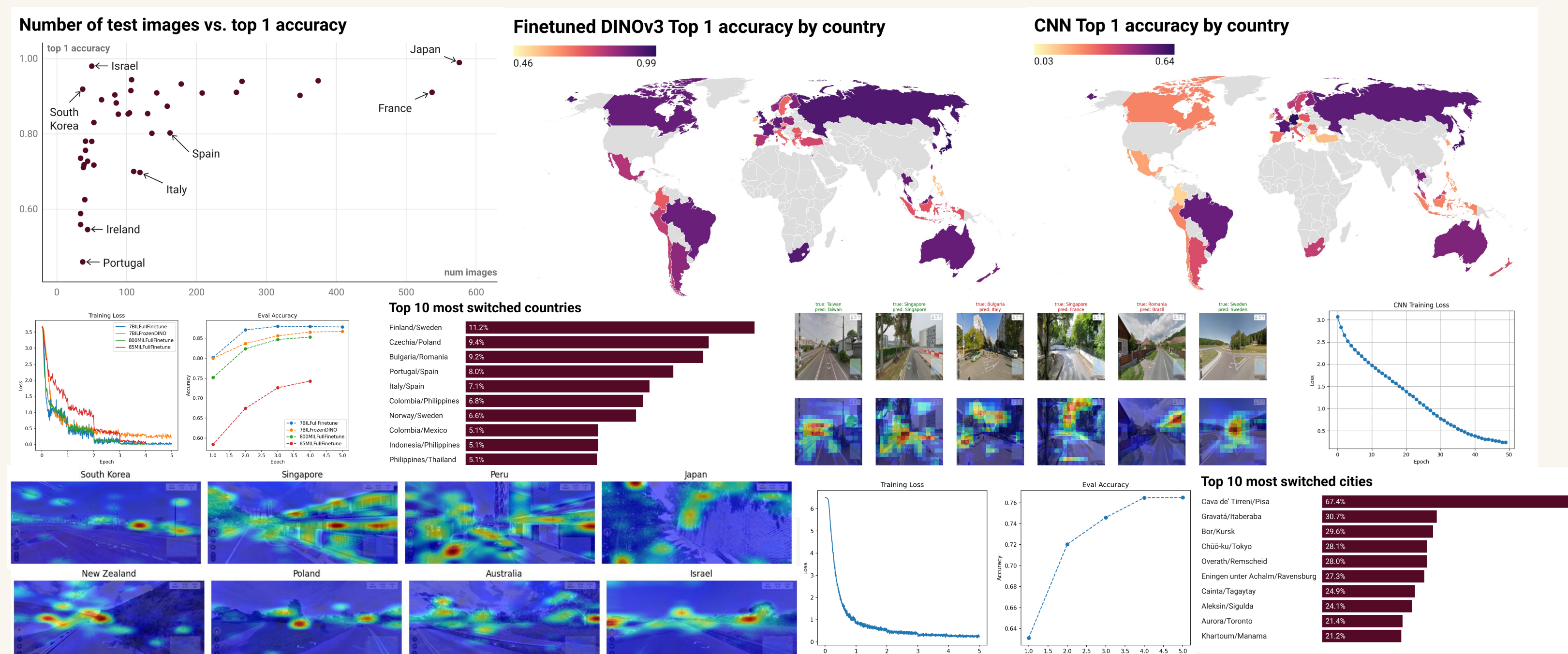
We trained the following models:

- 4 versions of finetuned/frozen DINOv3 on 39 countries
- Frozen DINOv3 on 1000 cities
- Small & large CNN
- Frozen DINOv3 coordinate
- Text recognition + identification
- SIFT-based model
- SIFT + K-means clustering

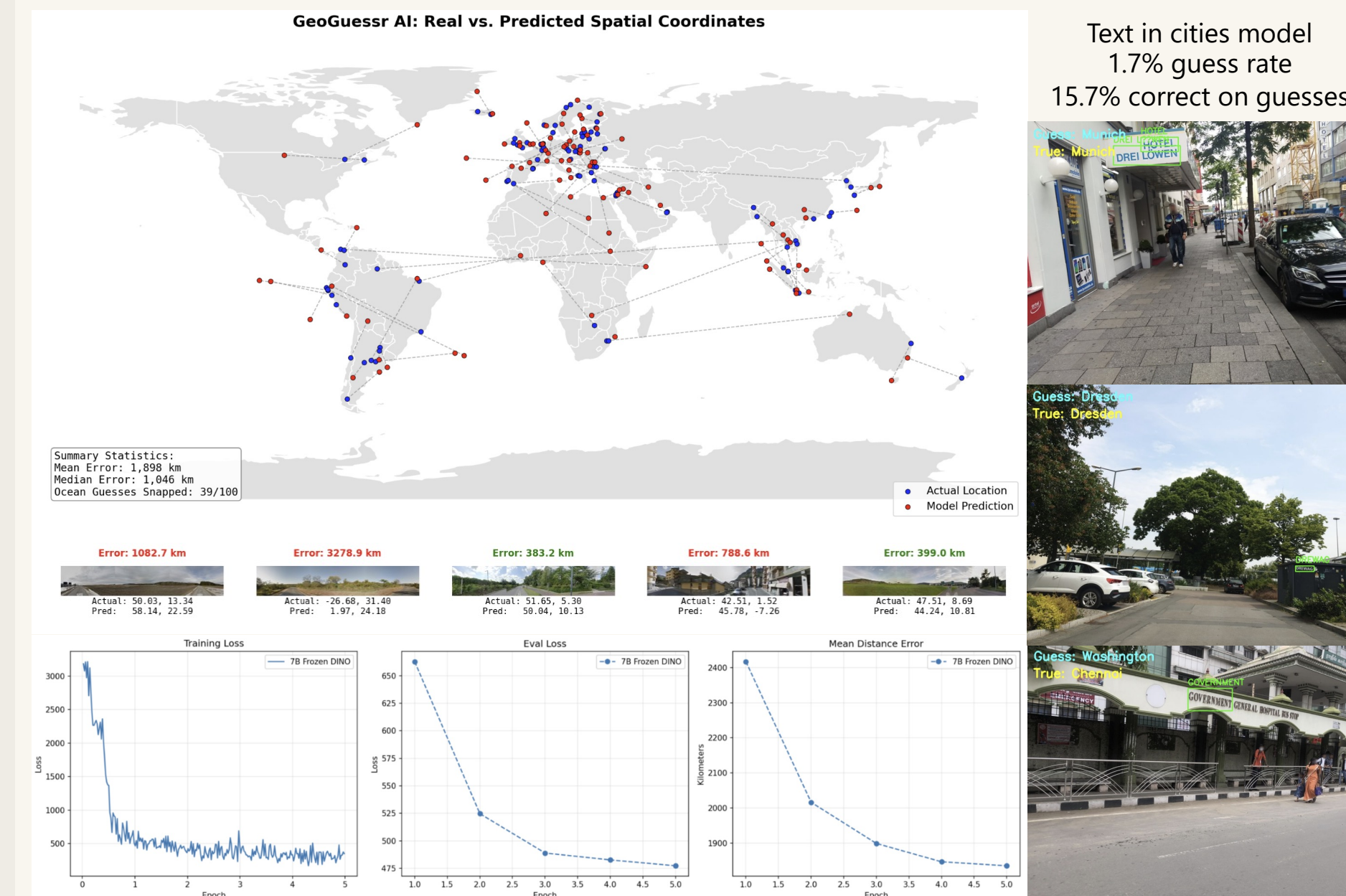
Model architectures



Results



More results



References

- [1] Haas, Lukas, et al. "Pigeon: Predicting image geolocations." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.
- [2] Siméoni, Oriane, et al. "Dinov3." *arXiv preprint arXiv:2508.10104* (2025).
- [3] Moreno, M. (n.d.). *marcelomoreno26/geoguessr · Datasets at Hugging Face*. Huggingface.co. Retrieved May 4, 2026, from <https://huggingface.co/datasets/marcelomoreno26/geoguessr>
- [4] Donnelly, C., Hadjiivanov, A., & Kalverla, P. (2025). *streetscapes* (v0.3.2). Zenodo. <https://doi.org/10.5281/zenodo.15606641>
- [5] Truong, W. (n.d.). *random_streetview_images_pano_v0.0.2*. Huggingface.co. Retrieved May 4, 2026, from https://huggingface.co/datasets/stochastic/random_streetview_images_pano_v0.0.2

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